18/11/24

enzymes

Enzymes :-

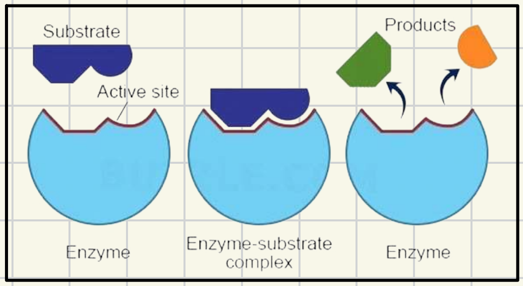
→ biological catalysts made of globular proteins

→ catalyse intracellular & extracellular reactions

e.g catalase inside liver cells or trypsin in the small intestine

→ the active site is specific & unique in shape due to the specific folding & bonding in the tertiary structure of the protein

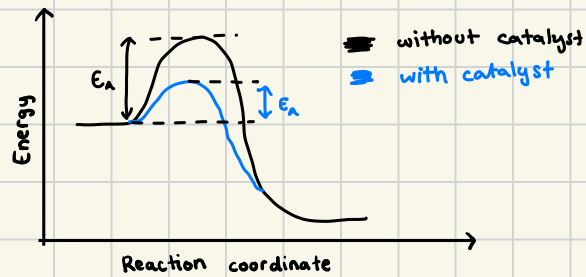
→ due to this, the enzymes can only attach to substrates that are complementary in shape



Activation energy:-

→ all reactions require a certain amount of energy before they occur, known as activation energy (E_A)

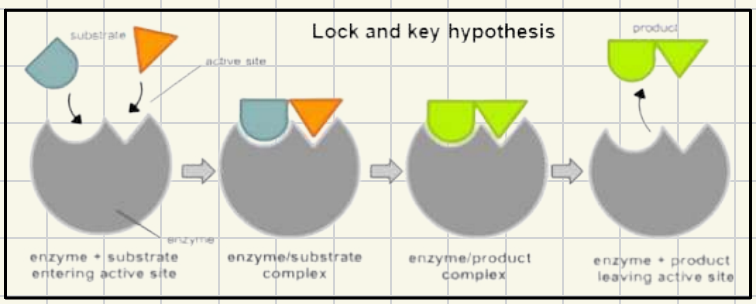
→ when enzymes attach to the substrate, they can lower the E_A needed for the reaction to occur & thus speed up the reaction.



Lock-and-key hypothesis

→ this model suggests that the enzyme is like a lock & the substrate is like a key

→ its thought that the active site is a fixed shape and that due to random collision, an enzyme-substrate complex



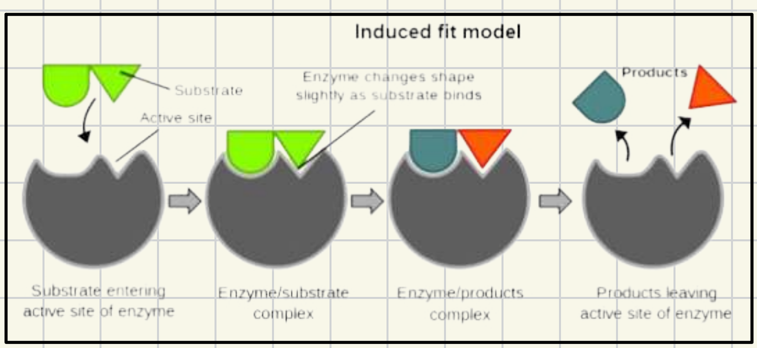
→ the charged groups within the active site are thought to distort the substrate thus lowering 'E_A'

Induced Fit hypothesis

→ this model suggests that the enzyme is like a glove & the substrate is like your hand

→ it's thought that the active site is induced, or slightly changes shape, to mould around the substrate

→ when the enzyme-substrate complex occurs, it puts strain on the bonds & thus lower 'E_A'



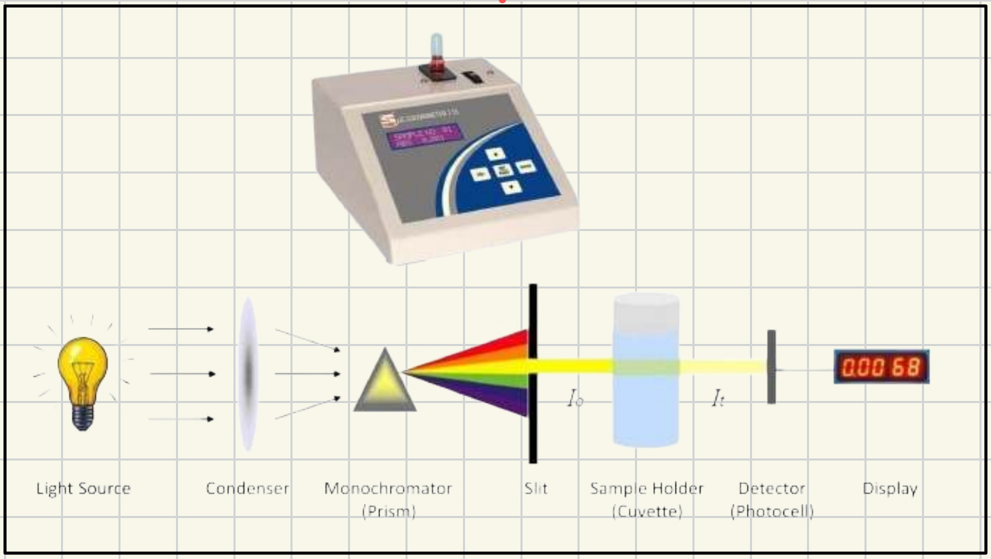
Investigating enzyme-catalysed reactions

- catalase reaction monitoring
  - catalase is an enzyme that catalyses the breakdown of hydrogen peroxide (substrate) into water & oxygen
  - to investigate the progress of this reaction, we can measure the rate of formation of products, namely oxygen, over time
  - this can be done by collecting the oxygen gas evolved during the reaction in a calibrated measuring cylinder & recording the volume at regular time intervals
  - alternatively, the rate of oxygen production can be indirectly measured by using a gas pressure sensor connected to a data logger, which records changes in pressure
  - by plotting the volume or pressure of oxygen produced against time on a graph, we can visualise the rate of the catalase-catalysed reaction
- amylase reaction monitoring
  - amylase is an enzyme that catalyses the hydrolysis of starch into simpler sugars
  - to investigate the progress of this reaction, we can measure the rate of dissappearance of the starch substrate over time
  - this can be done by taking samples of the reaction mixture at regular intervals & test for the presence of starch using an iodine solution
  - as the starch is broken down into simpler sugars, the iodine solution will change from blue-black to orange/brown, indicating the depletion of starch
  - by quantifying the intensity of the remaining color or lack, using a colorimeter, the rate of starch disappearance can be determined

→ an instrument used to measure the absorbance/transmittance of light by a solution

→ it works on the principle that certain substances absorb light at specific wavelengths, resulting in a change of intensity of the transmitted light

→ enzyme-catalysed reactions often involve color changes due to the formation of products or loss of substrates



Factors affecting enzyme-catalysed reactions

① temperature

→ if temp. is too low, there is insufficient kinetic energy for successful collisions

→ if temp. is too high, the enzymes denature; the tertiary structure alters due to bonds breaking and active site changes shape thus enzyme-substrate complexes cannot be formed

② pH

→ too high or low pH interferes with the charges in the amino acids in the active site which causes bonds to break & tertiary structure alters thus enzyme denatures

* all enzymes have different optimal temp. & pH that they work at

③ enzyme concentration

→ low enzyme conc. will result in a lower rate of reaction

→ increasing enzyme conc. will increase the rate of reaction as enzyme-substrate complexes are more likely to form. This would happen until the rate of reaction plateaus due to insufficient substrate present to bind with the large number of enzymes

④ substrate concentration

→ low substrate conc. will result in a lower rate of reaction as there will be fewer collisions between the enzyme & substrate

→ increasing substrate conc. will increase the rate of reaction as enzyme-substrate complexes are more likely to form. This would happen until the rate of reaction plateaus due to the fact that all the enzyme active sites would be in use

⑤ inhibitor concentration

competitive inhibitors

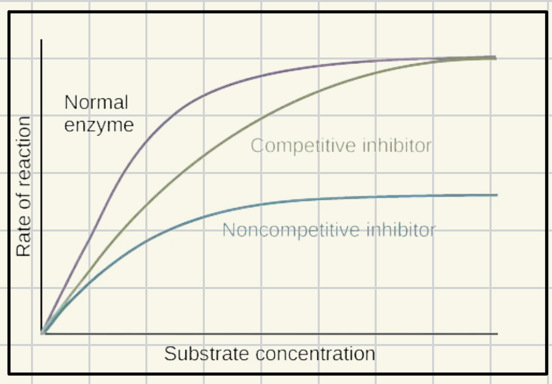
- complementary in shape to active site thus have the ability to bind to it
- this prevents the substrate from binding & enzyme-inhibitor complexes form instead of enzyme-substrate complexes.
- most competitive inhibitors are reversible so if a high enough conc. of substrate is added, the substrate can knock out the inhibitor & increase the rate of reaction

non-competitive inhibitors

- bind to the enzymes allosteric site, away from the active site
- this causes the active site to change shape and thus the substrate can no longer bind, regardless of how much is added
- as a result, fewer enzyme-substrate complexes form & the rate of reaction is much lower

* the products of some reactions are reversible inhibitors for the enzymes

* end-product inhibition enables the reactions to be controlled, to prevent overheating of cell or wastage of resources



Max. rate of reaction :-

→ V_{max} refers to the max. rate of reaction at which an enzyme catalyses a reaction

→ its reached when the active sites of all enzymes are fully saturated & all enzymes are working at max. capacity, so even if substrate conc. is increased, the rate of reaction wouldn't increase

→ V_{max} is a key measure to characterise enzyme kinetics & is determined experimentally by plotting reaction rate against substrate concentration until plateau is reached

Michaelis-Menten constant (Km) :-

→ Km is a measure of the affinity of an enzyme for its substrate. It represents the substrate concentration at which the reaction rate is equal to half of V_{max}

→ it can be thought of as the substrate conc. required to achieve half-maximal velocity

→ enzymes with lower Km values have a higher affinity for their substrates, as they reach half-maximal velocity at lower substrate conc.

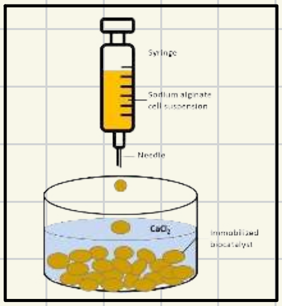
→ the Michaelis-Menten equation, which describes the relationship between reaction rate, substrate conc., V_{max} & V_{min}

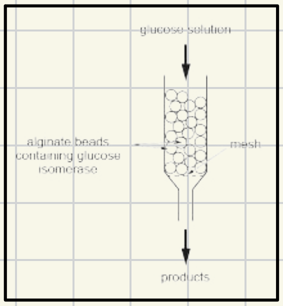
- Km values can be used to compare the affinity of different enzymes for their substrates
- enzymes with lower Km values exhibit greater substrate affinity & are more efficient at converting substrate into product, even at low substrate conc., while enzymes with higher Km values have lower substrate affinity & require more substrate conc. for maximal velocity.

Enzymes immobilised in alginate vs. in free solution

How to encapsulate enzymes in alginate beads?

1) mix enzymes with sodium alginate
2) add drop by drop into calcium chloride
3) pack them into a column & run it through
4) collect products from bottom





advantages of using immobilised enzymes

+ can easily reuse & recover the enzymes
+ longer shelf-life of the enzymes
+ less purification / downstream processing needed because product is enzyme-free
+ allow continous production & can obtain more product per unit time
+ reduces end product inhibition
+ enzymes more tolerant of pH changes & are thermostable